

The latent channel buys length, not answers

Qwen3-14B · AIME24 · $n=30$ paired · greedy

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Results

All five arms: same 30 items, same cap, same decode path.

arm	Judger input	KV	correct	capped	tokens
none	no cache	0 MB	21/30	8	—
single	no cache, no-latent prompt	0 MB	21/30	8	+6% ($p=0.54$)
shuf	<i>wrong</i> problem's cache	108 MB	19/30	4	−21% ($p=0.0009$)
evict64	64 of 770 positions	10 MB	22/30	3	−20% ($p=0.0125$)
real	full cache (published)	108 MB	22/30	3	− 33% ($p=0.0003$)

- **Accuracy: nothing.** Every paired interval contains zero.
- **Tokens: large and significant.** −33% median, 23 of 30 shorter.
- **Two thirds of that survives shuffling** the cache onto the wrong problem.
- **A 10× smaller cache scores the same** and keeps most of the saving.

The token effect is real, and mostly content-free

Exact sign test, paired, all 30 items.

step	isolates	shorter / longer	median
single $\rightarrow$ none	the prompt alone	10 / 14 ($p=0.54$)	+6%
none $\rightarrow$ shuf	<i>any</i> cache	23 / 5 ($p=0.0009$)	-21%
shuf $\rightarrow$ real	the <i>right</i> cache	19 / 9 ($p=0.087$)	-6%
none $\rightarrow$ real	published config	23 / 4 ($p=0.0003$)	-33%

- The **prompt contributes nothing**. The cache does all of it.
- -21 of the -33 needs only *a* cache, not the *right* one. Making it the right problem's cache adds -6%, $p = 0.087$.
- Mechanism is **termination**: runaway generations 8/30 $\rightarrow$ 3/30.
- Net win: upstream costs **0.9 s** against a **124 s** answer. End-to-end -37% wall clock, $p = 0.0001$.

9% of the bytes gets 59% of the saving

arm	KV	% of full	correct	capped	tokens vs none
none	0 MB	0%	21/30	8	—
evict64	10 MB	9.2%	22/30	3	− 19.6%
real	108 MB	100%	22/30	3	−33.0%

- Keeping **64 of 770 positions** by key-norm matches the full cache on accuracy *and* termination.
- **6.4× more saving per byte.** The curve saturates early; it is not graded in cache size.
- The dropped 91% still buys a further −8% ($p = 0.0081$) — real, but far less efficient per byte.

Where this lands

The channel is a conciseness scaffold, not an information channel. AIME is the third task to say so — MedQA and GSM8K already did (§9–§10).

Caveats.

- 30 items cannot resolve a one-problem effect. LatentMAS's own reported AIME24 gain *is* one problem; 10% needs 58 items, 5% needs 244.
- Greedy only, one model, one benchmark.
- shuf costs −3 items — and Frozen Mean-Replay independently costs the same −3 on AIME while matching Real on GSM8K/MATH. So content-independence holds for **length**, not for accuracy on the hardest items.

Also closed this week: steering the latent channel is a powered null (median −6%, $p = 0.87$, accuracy 22/30 → 22/30, 80% power for 16%).